

Inputs

Interfaces

Scientific core

Outputs

2D detector frames
and instrument headers

External 1D profiles
and correction metadata

SRM 3600, water, or
a user-supplied reference

CLI utilities and strict
BL19B2 batch workflow

SAXSabs Workbench
calibration and export

Python API
public package functions

Parse, normalize, and
validate inputs

Reduce detector data and
integrate with pyFAI

Estimate robust K and
propagate uncertainty

Check intensity state and
correction history

Write canSAS, NXcanSAS,
and calibrated 2D outputs

Absolute-scale 1D profiles
and calibrated 2D packages

canSAS XML and
NXcanSAS HDF5

Calibration records, source
hashes, QC, and run reports

Required checks before calibrated output: units · transmission · thickness · correction state · calibration provenance